

Predicting Icequakes and Benchmarking Quantum Reservoir Computing’s Viability

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June 2026

Abstract

Cryoseismic stick-slip events at the ice-bed interface are a type of icequake, serving not only as indicators of ice-stream dynamics with implications for sea level projections, but also as analogs for tectonic earthquake mechanics. Predicting the recurrence intervals of these events represents a challenging multivariate time-series prediction problem governed by complex, non-linear geophysical dynamics. Here, we evaluate whether quantum reservoir computing (QRC)—leveraging quantum entanglement and high dimensional state spaces—can improve forecasting for such complex geophysical systems. Utilizing an 11-year dataset of over 5,000 events from the Whillans Ice Plain (WIP) in Antarctica, we first establish a classical machine learning baseline for this temporal regression task. Our XGBoost model achieves a test mean absolute error (MAE) of approximately 2.83 hours, representing a ~ 4.2 -fold reduction in timing uncertainty over an observer without a model. We then apply the QRC framework to this dataset, and find that demonstrating that noiseless QRC simulations successfully extract meaningful temporal signals and achieve a marginal reduction in MAE over the classical baseline. To evaluate physical viability, we deploy these architectures on IBM’s 156-qubit Heron processor. The hardware results demonstrate

that shallow QRC designs are preferable to deep ones on noisy physical devices where hardware noise can overwhelm potential computational advantages. Finally, we discuss how the dataset’s dynamical characteristics may dictate the boundaries of this slight quantum advantage, offering perspectives on the broader application of quantum-recurrent frameworks in Earth systems.

1 Introduction

The forecasting of non-linear, chaotic, and multivariate dynamical systems remains one of the most enduring challenges in computational geophysics. While classical machine learning (ML) has made significant strides across the geosciences, emerging quantum computational frameworks offer entirely new paradigms for tracking complex dynamics. In particular, quantum reservoir computing (QRC) leverages the vast, high-dimensional Hilbert space of quantum mechanics to process complex time-series data without the prohibitive training costs of alternative variational quantum architectures. However, the deployment of QRC frameworks on noisy, real-world earth systems data remains an uncharted frontier which is just beginning to be explored [1], and the precise boundaries of its

applicability are an area of active research.

Cryoseismic stick-slip events at the ice-bed interface constitute an ideal natural laboratory for testing these emerging computational frameworks. Beyond serving as a scaled, accelerated analog for tectonic fault mechanics, this system is intrinsically critical to global climate projections and is uniquely supported by an exceptionally rich observational dataset. The rapid collapse of the West Antarctic Ice Sheet is estimated to contribute approximately 3.3 meters to global sea level rise, with ice streams serving as the primary conduit for mass transfer to the ocean [2]. Marine-terminating ice streams account for up to 90% of Antarctic ice sheet discharge [3], rendering their dynamics a first-order control on sea-level projections over decadal and centennial timescales. The Whillans Ice Plain (WIP), located at the downstream portion of the Whillans and Mercer ice streams flowing into the Ross Ice Shelf, is unique because its flow is dominated by macroscopic stick-slip cycles: sudden, 30-to-60 minute displacements of approximately 0.4 meters occurring once or twice per day, with negligible motion in between [4], [5], [6]. Releasing energy equivalent to a moment magnitude 7 earthquake, each slip event is detectable at distances exceeding 1,000 km via teleseismicity [7].

Slip events on WIP are tidally paced, occurring preferentially during falling tide under the governance of rate-and-state friction at the ice-bed interface [5], [7]. This stick-slip cycle reflects a delicate competition between tidal driving forces and basal friction, heavily modulated by subglacial hydrology [4], [8], [9]. A key hallmark of this system is the bimodal distribution of its recurrence intervals; slips occur either once or twice per tidal day, yielding distinct inter-event windows of roughly 12 or 24 hours [10], [11]. Whether a low-tide event nucleates in a given cycle depends on whether the instantaneous tidal stress can overcome a frictional threshold that evolves with subglacial water pressure and the physical state of the interface [8], [10]. This threshold sensitivity is further compounded by the long-term deceleration of WIP, which reduces the mean stressing rate and shifts the system closer to conditions where slip fails to nucleate entirely [12], [13]. Because nucleation depends on this non-linear interplay rather than tidal phase alone,

predicting event timing remains a notoriously complex forecasting problem.

To enable data-driven solutions, Katz et al. [6] compiled an 11-year (2008–2019) catalog of over 5,000 stick-slip events using a 48-station GNSS network. This record, which captures long-term tidal modulations and structural regime shifts, represents one of the longest, most complete continuous histories of ice-stream seismicity available, providing the rigorous baseline required to benchmark novel computational frameworks.

To establish a rigorous benchmark for this predictive task, we first leverage classical machine learning. Within the existing literature, machine learning has been increasingly applied to seismic event prediction—including aftershock forecasting and magnitude prediction in tectonic seismology [14], [15], as well as event detection and classification on WIP seismic records [16]. However, predicting continuous inter-event recurrence intervals remains unaddressed in cryoseismology. For structured tabular regression on datasets of this scale, gradient-boosted tree methods like XGBoost and CatBoost represent highly powerful, state-of-the-art classical baselines [17], [18], [19]. We therefore deploy these tree-based architectures to establish the first classical baseline for icequake timing regression, providing a standard against which to evaluate quantum alternatives.

Quantum reservoir computing (QRC) offers a distinct suite of properties that make it uniquely promising for this temporal regression task. By fixing the internal dynamics of the quantum reservoir and training only a classical linear readout framework [20], the number of trainable parameters remains independent of model depth and internal reservoir complexity—a significant architectural advantage when working with finite geophysical catalogs. Simultaneously, the quantum reservoir implicitly projects input features into a high-dimensional Hilbert space that scales exponentially with the number of qubits. This vast state space can potentially capture non-linear interactions and cross-correlations between multi-scale tidal forcing variables that classical regressors struggle to represent [21], [22]. Furthermore, because the quantum states are fixed rather than optimized

via gradient descent, QRC inherently avoids the "barren plateau" problem of vanishing gradients that severely limits deep alternative variational quantum architectures [23]. Although noiseless quantum kernels often span comparable function classes to classical methods [24], structured hardware noise possesses the unique potential to act as an algorithmic regularizer under specific conditions [25]. While QRC may have demonstrated early performance advantages on simulated chaotic data and physical hardware runs of specific financial time-series [1], [26], the true boundaries of quantum advantage on highly non-stationary, noisy, field-collected geophysical data remain fundamentally unestablished.

In this work, we bridge the gap between quantum machine learning and computational geophysics by deploying a QRC framework to predict cryoseismic recurrence intervals using the 11-year WIP dataset. We first establish a robust classical baseline using XGBoost to quantify the bounds of predictability using standard statistical methods. We then implement a QRC architecture in both noiseless simulations and directly on physical hardware using IBM’s 156-qubit Heron processor. Through this empirical benchmarking, we demonstrate a marginal quantum advantage in forecasting accuracy in QRC simulations, map the impacts of hardware noise on deep versus shallow reservoir architectures, and discuss how the underlying dynamical characteristics of Earth systems data dictate the boundaries of early quantum acceleration.

2 Methods

The goal is to predict the time-to-next-slip (TTNS) of the Whillans Ice Plane based on measured, physically relevant parameters of the system. The difference between the predicted TTNS and actual TTNS is measured as an absolute value (Mean Absolute Error - MAE) and a root mean square error (RMSE). Since RMSE is more sensitive to outliers, MAE is used as the primary metric for the prediction.

2.1 Data

The dataset of 5,150 stick-slip events over 11 years on WIP is taken from Katz et al., [6]. Six features were engineered from the raw data for modeling following previous work [27]. Each derived feature is physically motivated: four capture tidal state, and two capture slip history. This choice reflects the findings in previous work [6] of primary physical drivers of the slip events. All models operate on a windowed representation of 20 consecutive events, where the window length was selected empirically (as seen in Table 3) by sweeping the window size parameter for an XGBoost model and minimizing MAE. Thus, a 126-dimensional feature vector (21 events x 6 features per event) was obtained for input to the models.

2.2 Pre-processing and Baselines

A unified pre-processing pipeline was applied identically across all models prior to training. Events where the preceding or following inter-event interval was unknown were removed. The remaining events were randomly partitioned into training (60%), validation (20%), and test (20%) sets with a fixed random seed. Two naive baselines for TTNS prediction were implemented to assess whether learned models extract information beyond a simple training-set summary: a median baseline, which predicts the training-set median for every TTNS regardless of input, and a persistence baseline, which carries forward the most recently observed time between slips.

2.3 Classifier

Due to the bimodal inter-event interval distribution, the data were partitioned into two regimes corresponding roughly to the 12-hour versus 24-hour intervals present in the tidal data [6], before being routed to separate regressors. A diagram of the classifier architecture (section B) can be found in the Supplemental material. Cluster centers were identified via K-means with $k = 2$ fit exclusively on the training targets, giving a derived threshold of 64,875s corresponding to approximately 18 hours—thus sitting directly between the 12-hour and 24-hour subsets.

An XGBoost classifier was then trained on the raw input features to predict which regime a given event belongs to, using stratified cross-validation and class-weight balancing to account for the imbalance between short and long events. At the time of inference, the classifier assigns each event to the 12 or 24 hour regime using only its input features, never the actual TTNS. So, no label information crosses from training into predictions.

2.4 Classical ML Models

Three classical regressors were trained to establish the first machine-learning baseline in the literature for TTNS prediction at WIP. Each model receives the same windowed feature matrix, constructed by concatenating the six engineered features across the current event and its 20 preceding events, giving a 126-dimensional input (section 2.1).

Ridge regression provides a regularized linear baseline [28], with the regularization strength α selected by sweeping $\alpha \in [10^{-2}, 10^8]$ on a log scale and retaining the value that minimizes validation RMSE. XGBoost and CatBoost each build sequential ensembles of decision trees, differing primarily in how they handle variance across the training sequence [18], [29]. Both were tuned over 30 Optuna trials using the Tree-structured Parzen Estimator sampler, and optimized based on validation RMSE. The trial budget for XGBoost and CatBoost was kept deliberately modest to avoid fitting noise. To determine the highest-performing of the three models, the test MAE of each was compared.

2.5 Quantum Reservoir

A quantum reservoir encodes input data into a circuit, evolves it under a fixed Hamiltonian, and passes the resulting measurement statistics to a classical readout layer, in this case redridge regression [20]. Discovering whether this offers an advantage over classical feature extraction on real scientific data is one of the key aims of this study, and the design choices for the reservoir are intended to provide the best conditions for it to do so. Figure 7 shows the full quantum-classical pipeline, and trials of alternate reservoir architectures and their

corresponding MAEs can be found in Table 4 in the supplemental information.

Reservoir Hamiltonian: The Fully-Connected Transverse Field Ising (FC-TFI) and the nearest neighbor Transverse Field Ising (NN-TFI) models are chosen as the fixed Hamiltonians for the reservoir. They each offer distinct advantages. First, the all-to-all ZZ coupling of the FC-TFI weaves entanglement across every pair of qubits so that the reservoir’s internal state reflects correlations between input features. The NN-TFI’s ZZ couplings on the other hand enables information to flow through the system by coupling pairs of qubits while still keeping the number of two-qubit gates low to minimize noise effects on NISQ hardware. Second, the transverse field for both models drives quantum fluctuations that prevent the dynamics from settling into trivially classical behavior [30], [31]. Time evolution is approximated via Trotterization, as is normal practice, which alternates layers of ZZ interactions and single-qubit rotations.

Two such reservoirs are instantiated independently, one for each ‘class’ of slips (section 2.3), with Hamiltonian parameters drawn from different random seeds. A shared reservoir would produce correlated features for inputs that have physically different behaviors. Independent parameter draws ensure the two reservoirs develop distinct internal dynamics suited to their respective timescales [20], [32].

Input encoding : The encoding methods for a reservoir are critical in determining QRC performance. Here, we work with 6-qubit quantum circuits where each qubit maps to one of our physical features. The state of each feature for the 20 events in the window is encoded in the circuit using a Ry rotation gate. A handful of other encoding methods were tested [4] and can be found in the Supplemental Material. Notably, instead of summing rotations for the windowed events, each event has its own circuit thus receiving its own Ising parameters. This design choice was made for several reasons: first, it allows the model to differentiate between events. Second, we apply a classical exponential weighting of importance to the windowed events so that more recent events are given more weight, with weights $w_\ell = \exp(-\lambda\ell) / \sum_{k=0}^{20} \exp(-\lambda k)$, which is able to

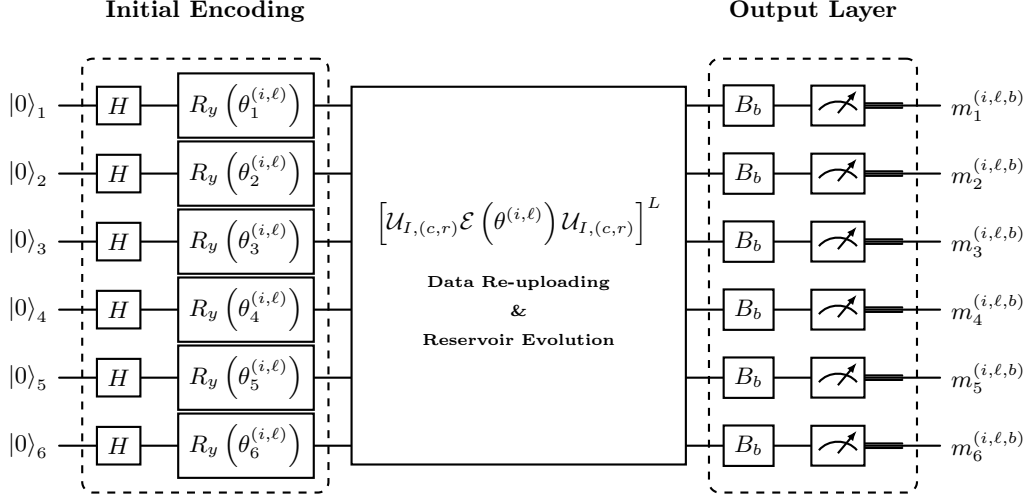


Figure 1: Quantum reservoir computing circuit and readout for sample i , event ℓ , and reservoir candidate c in regime r . The initial encoding layer applies a Hadamard gate to each of the 6 qubits, then maps the six selected icequake features of event ℓ to angles $\theta_j^{(i,\ell)}$ via R_y rotations, giving the joint encoding operation $\mathcal{E}(\theta^{(i,\ell)}) = \bigotimes_{j=1}^6 R_y(\theta_j^{(i,\ell)})$. This encoding is followed by L data re-uploading cycles, each consisting of the fixed Ising evolution $\mathcal{U}_I^{(c,r)}$ and a re-application of $\mathcal{E}(\theta^{(i,\ell)})$, without intermediate resetting of the quantum state. In the output layer, B_b rotates qubit j into the measurement basis $b \in \{X, Y, Z\}$, where $B_X = H$, $B_Y = HS^\dagger$, and $B_Z = I$, yielding outcome $m_j^{(i,\ell,b)}$. Single-qubit expectation values $\langle X \rangle$, $\langle Y \rangle$, $\langle Z \rangle$ and nearest-neighbor correlators $\langle Z_j Z_{j+} \rangle$ are computed from these measurements to form the 24-dimensional event representation $\phi_{c,r}^{(i,\ell)}$.

occur after the encoding since each event has its own circuit. Third, it allows for data re-uploading to occur between the windowed events. Data re-uploading re-introduces the input vector between successive layers of evolution, progressively building access to higher-order frequency components and enabling the circuit to approximate more nonlinear functions of the input [33]. This is a crucial feature of reservoir computing which separates it from extreme machine learning and ensures that the reservoir retains short-term memory and produces outputs that are affected by recent events. The weighted event representations are then concatenated to form $\Phi_c^{(i)} = (w_0 \phi_{c,r}^{(i,0)}, \dots, w_{20} \phi_{c,r}^{(i,20)}) \in \mathbb{R}^{504}$, which is passed to Ridge Regression to produce the per-candidate prediction $\hat{y}_{c,r} = f_{c,r}^{\text{Ridge}}(\Phi_c^{(i)})$. We find that this method of encoding gives us a $\sim 2,000$ second

improvement in TTNS when compared to encoding in 21 sequential rotations

Output layer: At readout, the reservoir state is projected onto single-qubit Pauli observables and nearest-neighbor ZZ correlators, where the nearest-neighbor index wraps as $j^+ = 1 + (j \bmod 6)$ so that qubit 6 couples back to qubit 1. The local observables capture individual qubit dynamics while the correlators expose the entanglement structure. These measurements are concatenated into a 24-dimensional event representation $\phi_{c,r}^{(i,\ell)} = (\langle Z_j \rangle, \langle X_j \rangle, \langle Y_j \rangle, \langle Z_j Z_{j^+} \rangle)_{j=1}^6 \in \mathbb{R}^{24}$.

Reservoir initialization: Because each reservoir is randomly initialized, any single instantiation may produce a less expressive feature space by chance. We run the full pipeline across multiple independent seeds and average predictions to ensure that any

observed parity or advantage reflects the approach itself rather than a fortunate or unfortunate parameter draw [34]. Due to time constraints on hardware, the reported MAE and RMSE for all QRCs, simulation and hardware, are the average of the top 3 ensemble iterations where the top 3 is selected by the highest MAE from the validation sets, giving the final prediction $\widehat{\text{TTNS}}^{(i)} = 1/N \cdot \sum_{n=1}^N \hat{y}_{c_n, r}$. Since our performance is evaluated on data we have not seen, this ensembling selection improves the prediction over a simple averaging of all 5 regressor results.

Additionally, the NN-TFI model is run both with the full dataset and with a 50% subset in noiseless and noisy simulations due to time constraints on quantum hardware. The full dataset simulations were run for direct performance comparison with the FC-TFI, and the subset was run in both simulations for direct hardware comparison—only the subset was able to be run on the quantum hardware.

2.6 Control Tests

The primary target variable of our predictive modeling framework is the time to next slip (TTNS). Multivariate time-series forecasting is an emerging area of investigation [1], [26], and continuous event-interval timing prediction remains significantly underexplored: [35] does it; however, they use a recurrence-free reservoir computing model rather than a standard QRC approach. To contextualize the performance of our framework within the broader QRC landscape, we executed control experiments using two alternative target variables derived from the same physical system: the instantaneous tidal state at the subsequent event, and the binary probability of a slip event occurring within the next 12 hours. Both benchmark configurations were deliberately selected to mirror data structures for which a quantum advantage has been empirically demonstrated in recent literature. Specifically, Hamhoum et al. [1] demonstrated a quantum advantage on the Lorenz-63 system and El Niño-Southern Oscillation (ENSO) data using a formulation where the input features represent the state of a system at time t and the output is the predicted state at a subsequent timestep.

To mirror this dynamical state-mapping architecture, we omitted the time-since-last-slip and slip-distance features from our catalog, isolating only the tidal variables. Using the tidal state of the current event as the input, the network was tasked with predicting the corresponding tidal parameters of the subsequent slip event. A notable divergence from the Lorenz and ENSO benchmarks is that the temporal sampling intervals between discrete states in our cryoseismic catalog are inherently non-uniform. Additionally, Ciceri et al. [26] demonstrated a hardware-based quantum advantage on a financial dataset utilizing a boolean classification target (whether a trade fill occurs). To emulate this binary classification task, we engineered a boolean target variable indicating whether or not a stick-slip event occurred within a forward 12-hour window, utilizing all six available features as inputs. For both control targets, the regime classifier was completely removed from both the classical and quantum pipelines, as partitioning the data into bimodal clusters would be physically meaningless for these alternative target structures.

2.7 Noisy Simulation

To model the effects of environmental decoherence, we executed noisy simulations of the quantum reservoir utilizing the Qiskit `AerSimulator` configured with the `FakeFez` noise backend, which provides an empirically grounded representation of the gate errors and readout noise characteristic of the `ibm_fez` hardware node. We ran a reduced simulation on only 6 qubits to maintain classical computational tractability, given the extreme overhead associated with simulating the full 156-qubit IBM Heron architecture upon which the physical hardware experiments were ultimately deployed. Conducting these noisy, lower-dimensional simulations enabled us to pre-optimize the model’s hyperparameters and circuit configurations under realistic noise constraints prior to hardware execution.

2.8 Hardware

To accelerate execution on the physical quantum processor, IBM’s 156 qubit Heron, we implemented an event parallelization scheme across the quantum

reservoir. 26 discrete events, each requiring a 6-qubit subspace, were concurrently encoded and run on the quantum reservoir. This was to take full advantage of the 156-qubits available to us. Since the reservoir utilizes Trotterized Ising interactions for state entanglement, the heavy-hex topology of the `ibm_fez` processor was natively compatible with our architecture, requiring no modifications to the underlying Ising reservoir dynamics. This allowed the independent 6-qubit sub-reservoirs for distinct events to be spatially multiplexed across the chip layout simultaneously. Readout remains similarly uncoupled, since extracting the expectation values of a parallelized event only requires measuring the specific 6-qubit register assigned to that event. In addition to this, the circuits were transpiled in `qiskit` for the `ibm_fez` backend, utilizing optimization level 3 to ensure maximum gate-count reduction and depth optimization prior to physical deployment.

3 Results and Discussion

3.1 Classical Model Performance

The full model performance results are reported in Table 1. To contextualize these metrics, two classifier-naive baselines (Section 2.2) were computed on the test set. The median baseline achieved an MAE of 17,168 s and a RMSE of 20,686 s, while the persistence baseline performed poorly by comparison, yielding an MAE of 22,803 s and a RMSE of 28,630 s.

The upstream regime classifier achieved an overall test accuracy of 76.5% in relation to the defined k -means boundary. Crucially, this classification accuracy establishes a strict upper bound on the performance of the two-stage pipeline: any event misrouted to the wrong regime regressor is necessarily evaluated against the wrong distribution, preventing the downstream model from recovering from the initial routing error. Thus, improving regime classification represents the highest-leverage direction for future work, potentially via the incorporation of continuous tidal forcing variables or richer proxies of the subglacial state.

Conditioned on this upstream classification step, all

Model	MAE (s)	RMSE (s)
Median baseline	17,168	20,686
Persistence baseline	22,803	28,630
Ridge Regression	9,823	17,280
CatBoost	9,215	17,045
XGBoost	9,098	17,129
FC-TFI (noiseless sim)	9,074	16,992
FC-TFI (noisy sim)	9,476	17,657
FC-TFI (hardware)	11,843	17,827
NN-TFI (noiseless sim)	9,124	17,087
NN-TFI (noisy sim)	9,243	17,243
50% NN-TFI (noiseless sim)	11,007	19,623
50% NN-TFI (noisy sim)	11,862	20,413
50% NN-TFI (hardware)	11,658	19,652

Table 1: Test set performance for all models. Baselines are classifier-naive. Best overall result in bold.

three classical regressors substantially outperformed both baselines, confirming that the 126-dimensional windowed tidal feature representation carries a robust predictive signal. XGBoost achieved the best overall performance across all metrics, with a test MAE of 9,098 s (approximately 2.53 hours), which represents a 47% reduction in timing uncertainty relative to the median baseline. CatBoost performed comparably, yielding an MAE of 9,229 s, while Ridge regression proved to be the weakest classical model with an MAE of 9,823 s.

The XGBoost test MAE of 9,098 s is approximately 2.5 hours. Because WIP stick-slip events occur once or twice per tidal day, an observer without a predictive model faces a baseline temporal uncertainty spanning 12 to 24 hours. Consequently, an average prediction error of 2.5 hours translates to a roughly 4.75- to 9-fold reduction in timing uncertainty, depending on the true underlying

distribution regime of the given cycle. This reduction is operationally significant; instead of requiring continuous instrument deployment, field-based target data collection can be strategically concentrated around the predicted temporal window, capturing the event with high probability while dramatically reducing field operational costs. Additionally, since icequake timing directly constrains the transient rate-and-state friction dynamics governing ice-stream velocity, establishing significantly tighter predictive bounds on these failure events provides a data-driven framework to advance our understanding of subglacial glaciological processes.

The performance gap between Ridge regression and the tree-based models, as shown in table 1 suggests that the mapping from tidal-state and slip-history features to inter-event interval is nonlinear, with gradient-boosted ensembles better suited to capturing regime-dependent interactions across the 126-dimensional feature space. The smaller margin between XGBoost and CatBoost suggests that the performance frontier for tree-based methods is near saturation with the current feature set, and that meaningful further classical improvements are likely to require either additional rich features or a more accurate upstream classifier.

3.2 QRC Simulation

3.2.1 Noiseless Simulation

The noiseless FC-TFI quantum reservoir simulation ensemble with TTNS as the target achieved a MAE of 9,054 s, and a RMSE of 16,965 s. Individual iterations varied in MAE from 9,080 s to 9,177 s. Within the ensemble, short-regime MAE ranged from 8,495 s to 8,561 s across iterations, and long-regime MAE ranged from 15,071 s to 15,177 s. Figure 2 (b) shows predicted versus true time-to-next slip (TTNS). The colors of the points indicate their distance from the predicted regime’s K-means centroid—how similar that data point is to the typical data points in that subset (either 12-hour subset or 24-hour subset). Points close to the centroid (dark) reflect high-confidence, regime-typical predictions, while brighter points lie further from it. The short-interval cluster

aligns closely with the diagonal line (representing a perfect prediction) and has a low centroid distance, corresponding to its MAE of 6,478 s. However, the long-interval cluster is more dispersed, with higher centroid distances, reflecting greater prediction uncertainty at longer recurrence intervals, resulting in a MAE of 15,215 s. Furthermore, looking at the subset of high-confidence samples—where the classifier was 80% confident of the regime to which the event belonged—accounted for 281 of the events, which is approximately 49.7 % of the test set, and gave a MAE of 5,226 s. This nearly 50% reduction in error further indicates that the pipeline’s primary bottleneck is the classifier’s accuracy. Further tests were conducted to assess how the QRC performed compared to classical methods under different input conditions, such as a smaller subset or noisier features, and the noiseless QRC simulation consistently performed approximately 100 seconds better than classical XGBoost. The full results can be found in Appendix C of the supplemental material D.

Crucially, the noiseless QRC architecture achieved an MAE of 9,074 s, performing on par with the heavily optimized classical XGBoost baseline (9,098 s). This parity is remarkable given that QRC is an emerging, unoptimized paradigm that remains entirely untested for cryoseismic regression tasks, whereas gradient-boosted trees represent a mature, highly refined apex for tabular machine learning. Demonstrating that an un-tuned quantum reservoir can immediately match the best classical architecture on this complex geophysical task under idealized conditions suggests that for this specific windowed, tabular feature engineering scheme, both architectures have converged on the maximum available predictive signal. Furthermore, because the reservoir successfully extracts high-dimensional predictive signals directly from noisy, field-collected earth systems data, this work establishes QRC as a highly viable frontier framework for computational geophysics. These findings strongly imply that quantum reservoir architectures merit deeper exploration across diverse geoscientific domains, where a distinct quantum advantage may readily emerge on datasets characterized by different structural feature distributions, higher dimensionalities, or continuous

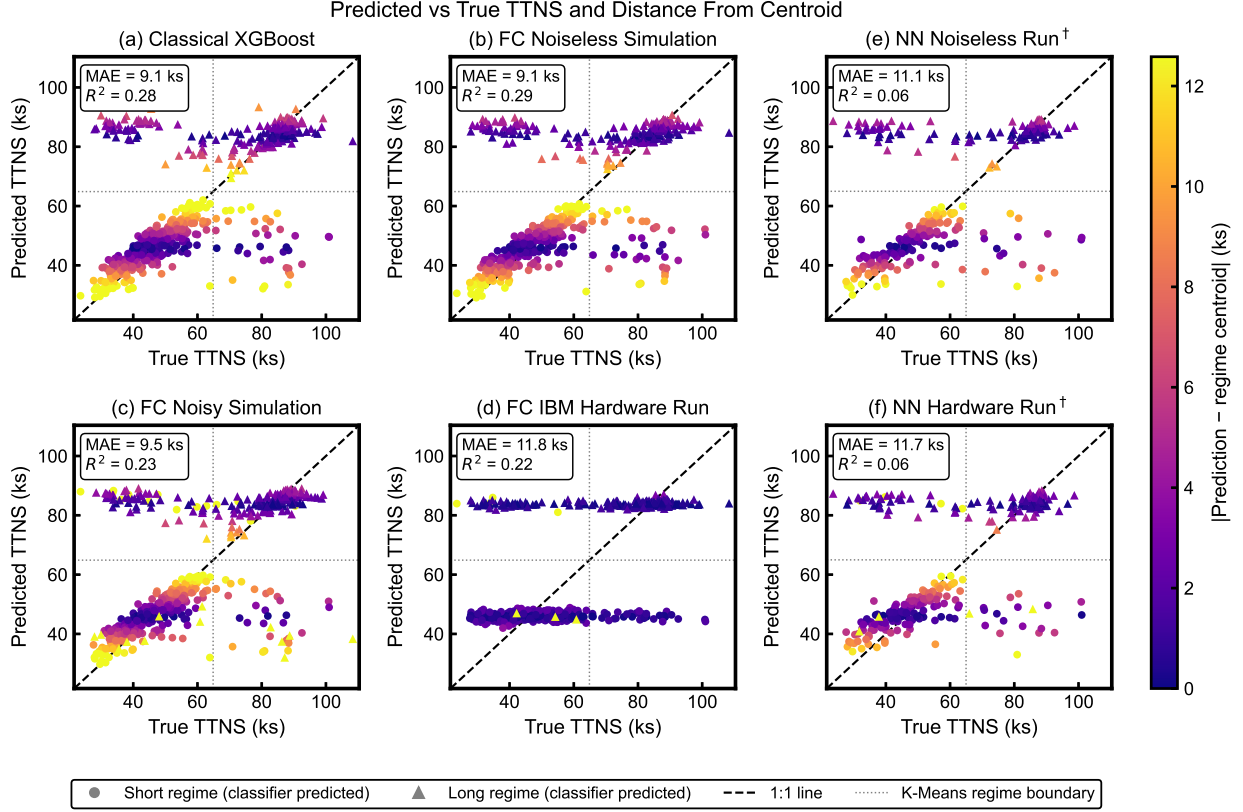


Figure 2: True versus predicted plots for FC-TFI Hamiltonian (a) Classical XGBoost, (b) Noiseless simulation, (c) FC-TFI Noisy simulation on **FakeFez** noise backend, (d) FC-TFI hardware results on the **ibm_fez** quantum compute node, (e) NN-TFI Noiseless simulation, and (f) NN-TFI hardware results. The color bar represents distance from the centroid, and the circles and triangles denote short versus long regime values. † means that only 50% of data was used, due to resource constraints.

multi-scale dynamics.

Similarly, the NN-TFI noiseless simulation with the full dataset achieved comparable results to the FC-TFI with a MAE of 9,124 and a RMSE of 17,087. The consistency between these two structures is notable for QRC architectural design: the connectivity between qubits can be reduced from long-range to local coupling without significantly degrading the reservoir’s performance. The NN-TFI with a 50% subset achieved a MAE of 11,007 and RMSE of 19,623.

3.2.2 Noisy Simulation

As shown in table 1, the MAE error recorded for the FC-TFI in noisy simulation was 9476s, and it gave a RMSE of 17,657s. Individual iterations varied in MAE from 9504s to 9590s. Within the ensemble, short-regime MAE ranged from 6768s to 7050s across iterations, and long-regime MAE ranged from 13,842s to 14,077s. As in noiseless simulation, the NN-TFI in noisy simulation with the full dataset was comparable to the FC-TFI, giving a MAE of 9243s and a RMSE of 17,243s. Individual iterations varied from 9280s

to 9243 s with the short-regime MAE ranging from to 7,025 s to 7,165 s and the long-regime ranging from 13,489 s to 13,574 s. With 50% of the data the MAE was 11,862 and the RMSE was 20,413.

The results of the noisy simulations suggest that quantum noise being injected into the system hurts the performance of the reservoir. The NN-TFI performing slightly better than the FC-TFI supports this analysis since a reservoir with lower connectivity has fewer operations for noise to degrade the performance.

3.2.3 Alternate Prediction Targets

To evaluate whether QRC properties could demonstrate advantage under alternative objective formulations, we benchmarked the QRC framework against the two control targets (section 2.6). For the first control task—predicting the multi-variable tidal state of the system at the subsequent slip event given the current event state—the noiseless QRC underperformed relative to the classical XGBoost baseline. The MAEs for this multivariate trajectory prediction can be found in Table 2. For the second control task—predicting the binary probability of whether a stick-slip event would occur within a forward 12-hour window—the noiseless QRC architecture achieved performance parity with the classical baseline, with both frameworks yielding an identical MAE of 0.32.

Tidal Feature	Classical MAE	QRC MAE
Tide Derivative	0.0921	0.0958
Form Factor	1.2384	1.1281
High Tide Event	0.4217	0.4221
Tide Height	31.9274	33.0489

Table 2: QRC performance on predicting the next tidal state given the previous tidal state.

These results suggest that the reason [1], [26] saw quantum advantage, while this advantage was not realized on the icequake dataset, is not explicitly tied to the nature of the prediction target.

3.3 QRC Hardware

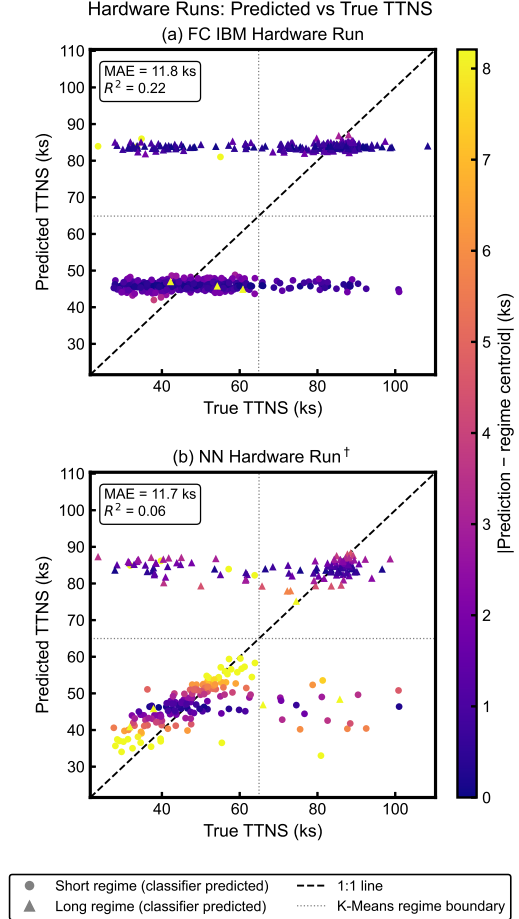


Figure 3: Hardware QRC Results

On IBM hardware, the ensemble achieved a test MAE of 11,843 seconds, a degradation of 2,769 seconds relative to the noiseless simulation that gave a MAE of 9,074 seconds. Panel d of Figure 2 shows how predictions collapse into two horizontal bands clustered near the regime means rather than tracking the continuous variation seen in simulation, suggesting the model has reduced to predicting the regime mean.

This collapse can primarily be attributed to noise accumulation across circuit depth, as the FC-TFI architecture generates on average a two-qubit depth

of 840 gates, and a total two-qubit gate count of 18,645 for a 6-qubit reservoir with a 20-event look-back window and data re-uploading. Using the error-per-layered gate metric $EPLG = 1 - LF^{1/n_2q}$ [36] the corresponding total circuit fidelity is approximately 2%. Given the hardware output is roughly 98% noise, the T_2 dephasing and coherent error accumulation present on the IBM Heron R2 device [37] drive the reservoir towards a maximally mixed state, and the classical regressor downstream receives no meaningful signal from the reservoir, which is the most likely reason it collapses to predicting the regime mean. Notably, some prior QRC studies have reported that mild hardware noise can act as a beneficial perturbation, effectively augmenting the reservoir’s expressivity and providing quantum advantage not seen in noiseless or noisy simulations [1], [25], [26]. However, at the circuit depths required for this architecture, the 98% noise floor is far beyond the regime where such effects are observed, and the potential noise advantage is suppressed. While emerging literature suggests error mitigation techniques applied during hardware execution [38], [39], [40], these were not explored in this project due to hardware time constraints. Additionally, the post-processing techniques seen in [38] were not implemented, as those do not target the structured, coherent noise that was dominant in our hardware run.

This observation is supported by the results of the NN-TFI 50% subset run. This architecture had a median two-qubit gate depth of 166 and a median two-qubit gate count of 1751, which led to the output being 55.6% error rather than the 98% error seen in the FC-TFI. With this noise floor, the NN-TFI architecture was able to achieve a MAE of 11,658 and a RMSE of 19,652. These error rates in addition to the error distribution seen in figure 2 show that a shallower architecture with less noise compounding in the system was able to extract useful signal from the given features.

In line with the noisy simulation results, having less connectivity allowed less opportunity for noise to degrade the QRCs’ performance. Furthermore, the NN-TFI on IBM’s Heron 2 chip specifically has less noise with the NN-TFI due to its topology: the long-

range interactions of the FC-TFI require a greater use of SWAP gates as compared to the NN-TFI. Thus, it is not surprising that the NN-TFI outperformed the FC-TFI.

These hardware results, in addition to the noisy simulation results, emphasize that shallower architectures with fewer long-range interactions are more suitable on NISQ devices than deeper, more connected architectures on datasets where noise has been shown to hurt rather than help performance. The architecture with less connectivity allows for less noise to be in the system, and its quantum properties achieve comparable performance to the QRCs with greater connectivity, as seen in the noiseless simulations. Furthermore, since the 50% NN-TFI still did not outperform either of the 50% simulations, this further shows that noise is not a useful regularizer to this dataset.

3.4 Overall Analysis of Results

When evaluating the underlying signal dynamics of our cryoseismic catalog against datasets that have historically demonstrated a quantum advantage, a critical nuance emerges regarding the specific manifestation of chaotic and stochastic behaviors. Clean, low-dimensional chaotic systems—such as the Lorenz-63 attractor—demonstrate the strongest performance gains in noiseless QRC simulations but fail to retain that advantage on physical hardware [1]. Conversely, highly stochastic, non-chaotic data—such as financial transaction fill probabilities—realize a distinct performance advantage on physical hardware rather than in noiseless simulation [26]. Macro-climate phenomena like the ENSO data possess mixed dynamical characteristics, showing a bifurcated performance profile: they outperform generic classical models in noiseless simulation while simultaneously benefiting from structured hardware noise [1]. This collective pattern indicates that the precise topological and statistical dynamics of the input signal dictate both the viability and the medium (simulation vs. hardware) of achievable quantum acceleration. Chaotic dynamics may leverage the vast, clean state-space mapping of a noiseless quantum reservoir, whereas stochastic systems can exploit

physical hardware noise as a beneficial regularizer, a mechanism supported by foundational studies on noise-induced regularization [25] and chaotic dynamic learning in QRC environments [41].

While an explicit characterization of the icequake system has not been conducted, the literature for analogous systems, namely slow-slip earthquakes [42], [43], [44] suggests that it is chaotic. Additionally, the WIP stick-slip cycle has been successfully described using rate-and-state friction, a nonlinear deterministic framework driven by tidal forcing [4], [6], [8], suggesting it is reasonable to treat the system as nonlinear and deterministic. This hypothesized characterization aligns with our above observations: the chaotic nature allows the noiseless QRC advantage to be marginally realized, while the hardware advantage is not realized due to its lack of stochasticity, even on the NN-TFI, which had a meaningful reduction in noise compared to the FC-TFI. A systematic analysis of QRC performance—spanning both noiseless simulations and physical hardware implementations across a broader spectrum of stochastic, deterministic, chaotic, non-chaotic, clean, noisy, continuous, and event-driven systems—is critically needed to map the exact boundaries of quantum advantage in the geosciences.

4 Conclusion

The WIP is a tidally paced West Antarctic ice stream whose movement is dominated by discrete stick-slip events (icequakes) occurring on a bimodal 12- or 24-hour cycle. Predicting exact inter-event timing remains a highly non-trivial forecasting problem due to the non-linear interactions among tidal forcing, basal friction, and subglacial hydrology that govern slip nucleation. Utilizing an 11-year GNSS stick-slip catalog compiled by Katz et al. [6], this study establishes the first machine learning and quantum reservoir computing application in the literature dedicated to cryoseismic inter-event interval regression.

Our two-stage classical pipeline—comprising an upstream XGBoost classifier that routes events into

two independent regime regressors conditioned on a 20-event history window—achieved a test MAE of 9,098 s (approximately 2.5 hours). This represents a 47% reduction in error relative to the median baseline and a 4.75- to 9-fold reduction in timing uncertainty relative to an observer with no model. Operationally, this predictive window enables highly concentrated, episodic field data collection within targeted intervals rather than requiring continuous instrument deployment across a massive 12- to 24-hour uncertainty window. Furthermore, tighter constraints on event timing translate directly into higher-fidelity observations of the slip cycle itself, which are essential for tracking how these dynamics evolve as the WIP undergoes long-term deceleration.

A central contribution of this work was evaluating whether QRC can offer a performance edge on this noisy, multivariate geophysical dataset. Remarkably, our noiseless QRC simulation achieved a test MAE of 9,074 s, marginally outperforming the heavily optimized classical XGBoost baseline. We highlight this parity as a major milestone: an emerging, entirely unoptimized quantum paradigm has demonstrated the capacity to match a mature, state-of-the-art classical benchmark on real-world earth systems data under idealized conditions.

Furthermore, the results of both simulations and the hardware runs, as compared to the results on other datasets described in the literature, reveal an important pattern between quantum advantage and the datasets dynamical characteristics: quantum advantage derived from hardware noise acts as a beneficial regulator to stochastic data [26]; however, this noise seemingly degrades the performance of non-stochastic datasets such as the Lorenz-63 [1] and in our icequake system which saw an approximately 400 s higher MAE in noisy simulation and a 650 s difference in the best hardware run as compared to the corresponding noiseless simulation (50% NN-TFI). Furthermore, chaotic systems, such as the Lorenz-63 and the WIP icequakes, achieve an advantage in noiseless simulation. Future work would benefit from a systematic evaluation of quantum advantage for datasets with varying dynamical characteristics.

Additionally, when deployed on IBM’s 156-qubit Heron R2 hardware (`ibm_fez`), device noise in

the fully connected (FC-TFI) reservoir compounded catastrophically due to deep circuit depths and extensive gate routing. This caused a complete signal-to-noise collapse, rendering the quantum readout states roughly 98% noise and leaving the downstream regressors only capable of predicting the distribution mean. To mitigate this hardware-induced decoherence, we implemented a shallower, nearest-neighbor (NN-TFI) reservoir architecture. By restricting entanglement topology, the NN-TFI reduced the hardware noise envelope to approximately 55.6%, allowing the downstream regressors to successfully extract meaningful physical signal and yield a hardware-run MAE of [to be filled in]. Notably, the NN-TFI performed comparably to the FC-TFI in noiseless simulations and substantially outperformed it under noise injection. This architectural divergence strongly implies that low-connectivity, hardware-native topologies are fundamentally more resilient on NISQ-era devices.

5 Acknowledgments

We acknowledge the use of time provided by IBM on their quantum computer for this work. The views expressed are those of the authors, and do not reflect the official policy or position of IBM or the IBM Quantum team.

Noah Braunfeld was supported through the National Science Foundation’s National Research Traineeship program under Grant No. 2125899.

We would like to thank Matthew Siegfried, Zachary Katz, and Ryan Venturelli for insightful conversations about the Whillans Ice Plain in particular, and geology in general, that helped inform the direction of this project. We thank Roberta Zambrini, Eran Ginossar and Mushin Tamturk for insights about quantum reservoir computing and its use in predictions based on tabular data.

This work used Alpine at the University of Colorado Boulder Research Computing through allocation `rmacc-general` from the Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS) program, which is supported by U.S. National Science Foundation

grants No. 2138259, No. 2138286, No. 2138307, No. 2137603, and No. 2138296.

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Supplemental Material

A Windowed Events

Lookback (prior events)	Validation MAE (s)
10	12,188
15	12,192
19	11,583
20	11,353
21	11,777
30	11,579
40	12,762
50	13,790

Table 3: Validation MAE as a function of lookback window length (prior events only, excluding the current event). A lookback of 20 prior events was selected for all models.

B Classifier Architecture

Train, test, and val refer to the training, testing, and validation subsets of the data. They are split during pre-processing and remain separate downstream. The y values of each subset represent the time-to-next-slip values, and the x values are the six features for each event. The x values are known at the time of prediction, and the y values are what are being predicted. A key design constraint of the classifier is that it routes events using only input features, never the target interval. The K-means threshold was derived exclusively from training-set targets, and at inference time, the classifier assigns regimes from features alone. Label information, therefore, cannot propagate from training into test-set routing. Consequently, the performance gains from having the classifier reflect better accounting for the bi-modalities of the events rather than from information leakage.

B.1 Regime Classifier Performance

As seen in Figure B.1, the classifier enables the subsequent independent regressors to make better predictions in accordance with the distribution of the data’s true values. The classifier achieved an overall test accuracy of 76.5%, in relation to the defined k-means boundary, and improved the test MAE from 11,383 seconds to 10,220 seconds, an error reduction of 10.2% as compared to the purely classical XGBoost regressor. Furthermore, the MAEs of the separate regime classifiers also differ: with the short regime regressor giving an MAE of 6,472 seconds and the long regime regressor giving an MAE of 15,485 seconds.

The per-regime MAE asymmetry. Is worth noting: the short-interval regressor achieves a MAE of 6,472 s while the long-interval regressor achieves 15,485 s, a factor of 2.4 difference. The long-interval regime is both rarer in the training data and inherently more variable, which reflects the threshold-sensitive nucleation dynamics that govern whether a second daily slip will occur, making both classification and regression harder for those events.

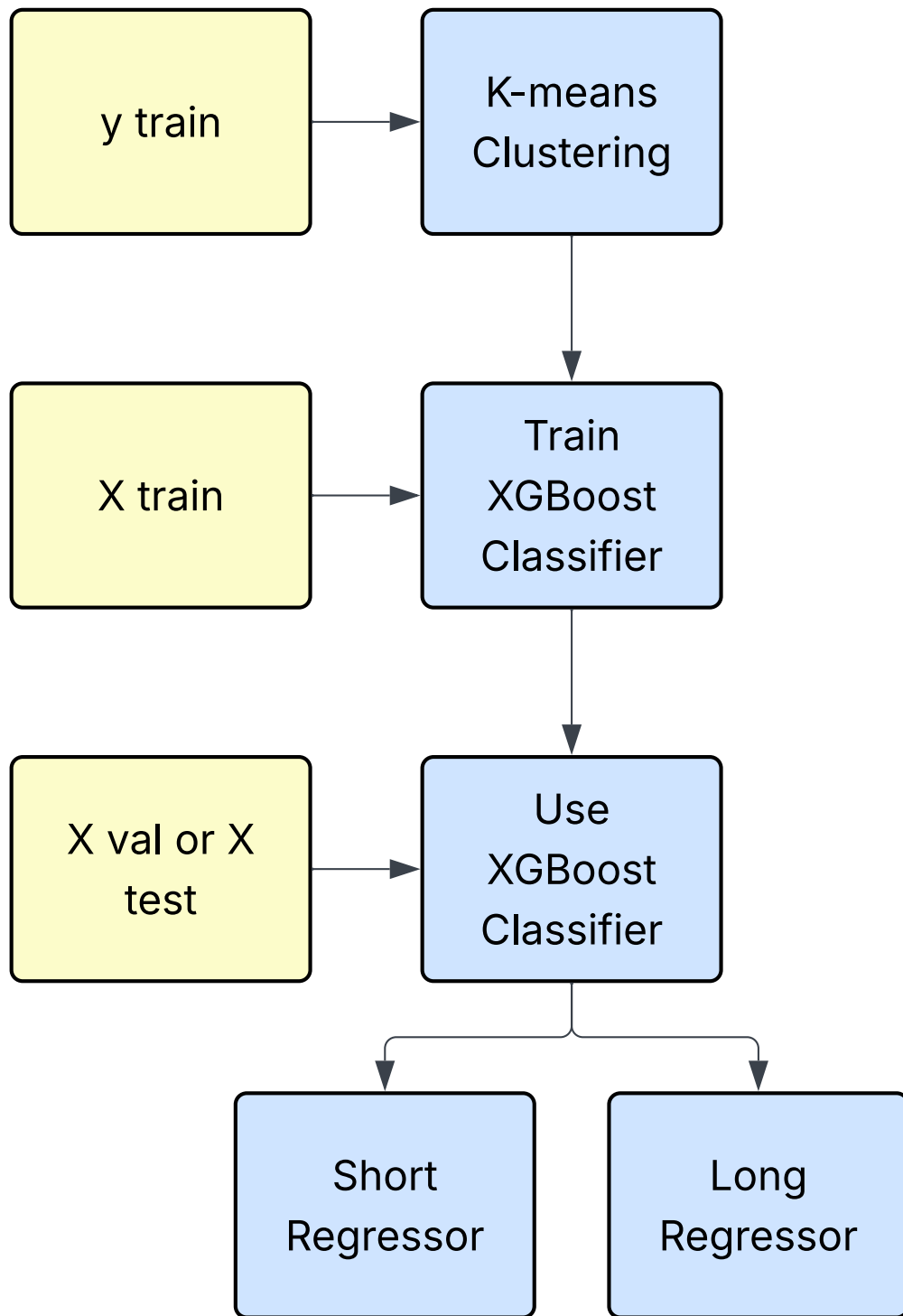


Figure 4: Classifier pipeline

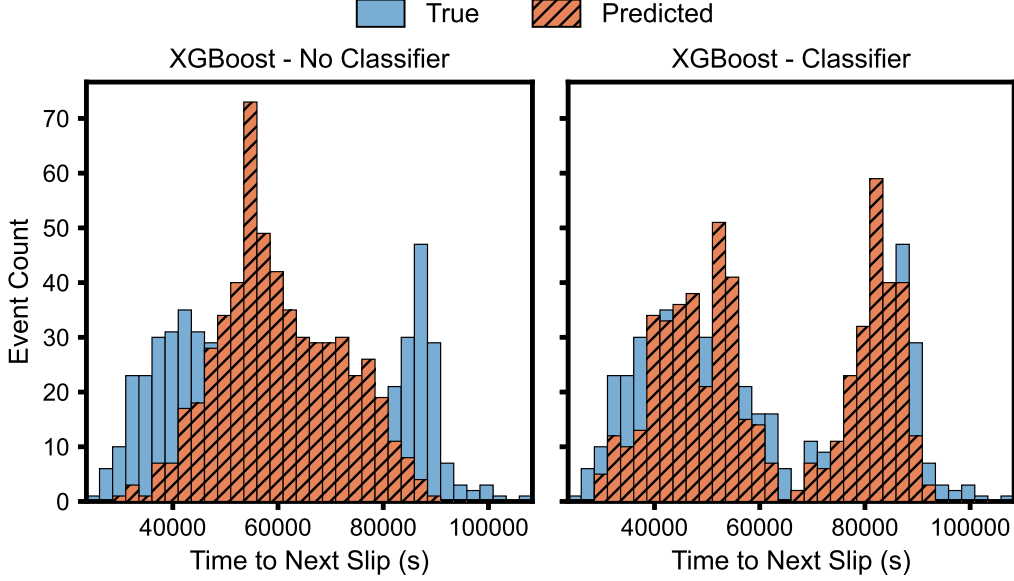


Figure 5: Histogram showing XGBoost performance before and after adding the classifier and subsequent time-constrained prediction.

C Alternate QRC Architectures

QRC variants were evaluated against a classical XGBoost and one other using the same 20-event look-back window. Unless otherwise noted, all QRC variants use angle encoding and single-qubit Pauli-X/Y/Z expectation values plus nearest-neighbor ZZ correlators as observables. Dense angle encoding maps two features per qubit via combined R_Y/R_z rotations rather than one. Instantaneous Quantum Polynomial encoding (IQP) encoding uses a Hadamard-diagonal-Hadamard feature map [22]. Multi-basis encoding measures observables across all three Pauli bases [30]. The recurrence-free variant removes inter-timestep quantum state propagation, functioning as a quantum extreme learning machine [34], [45]. The 5 look-back window was conducted to see if the 20 previous events were adding more noise than a predictive signal in the reservoir.

D Testing QRC vs Classical with Different Data Conditions

This figure shows how the QRC performs as compared to classical with different input conditions in the data. The first one is the regular one for reference, the next two show noise being injected into the slip distance feature of the data set, with the unstructured noise being applied through an additive Gaussian noise model and the structured noise being added through a log-normal multiplicative noise model. Finally, fewer features represent taking the top-most important value for each feature from the past 20 events based on calculated Shapley values and dropping the rest.

Model	MAE	RMSE
Classical XGBoost (with 20 prior events)	11,353	15,157
QRC with data reuploading, one circuit per event (21 per event)	11,678	15,119
QRC with only 5 look-back events	12,229	16,418
QRC with Pauli measurements	11,678	15,119
QRC with dense angle encoding	12,827	16,150
QRC with IQP encoding	12,563	15,893
QRC with multi-basis encoding	11,903	15,456
Recurrence-free QRC (echo state network)	11,682	15,132
QRC with Ising Hamiltonian evolution	11,516	14,912
QRC with Hamiltonian evolution and data reuploading	11,492	15,011

Table 4: Test set performance across all models. Best QRC result in bold.

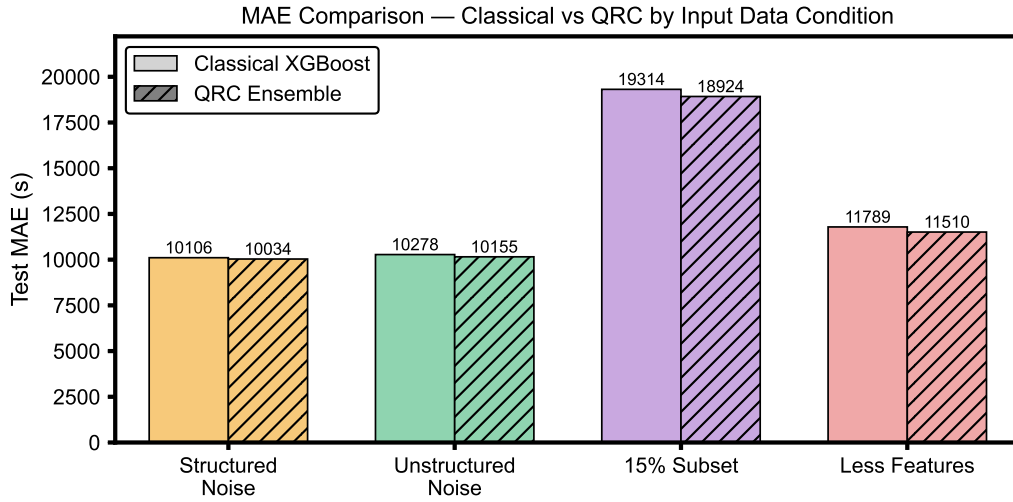


Figure 6: MAE classical vs QRC comparison for different input data conditions.

E QRC Pipeline

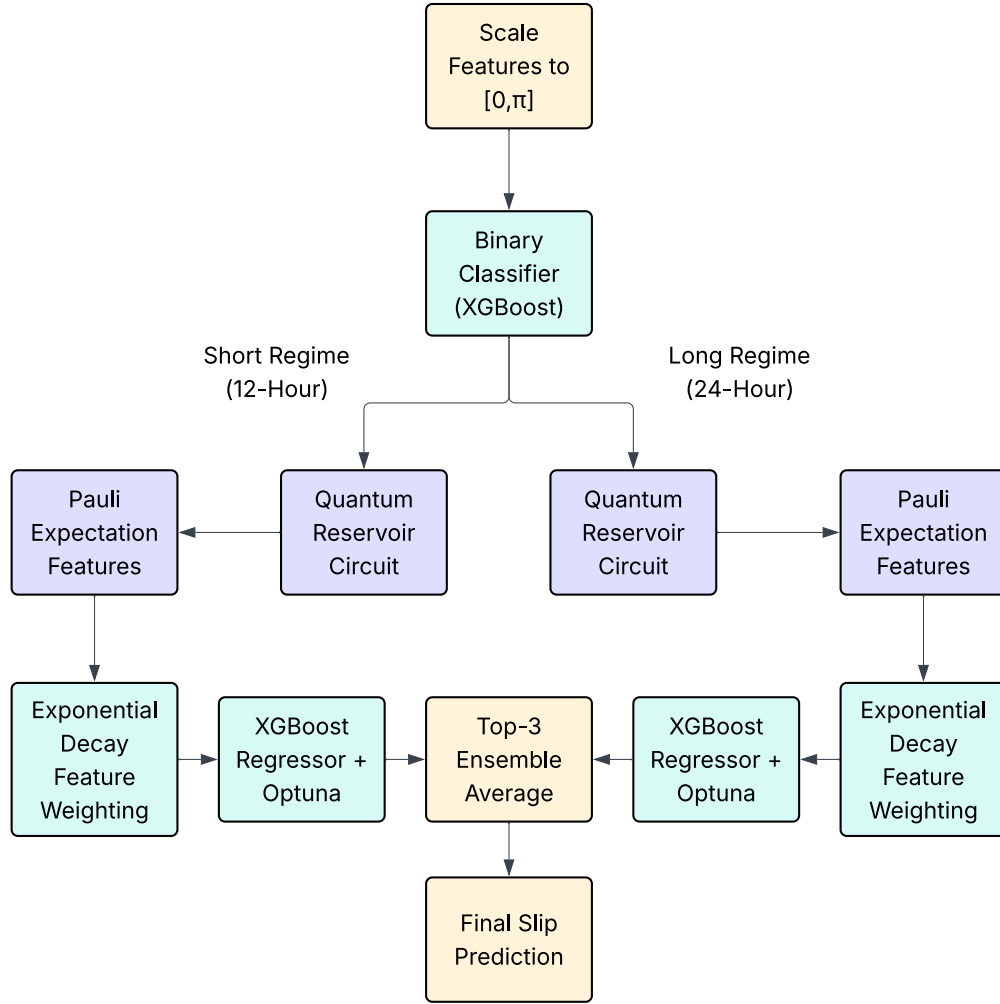


Figure 7: Complete quantum reservoir pipeline.